ligase~ — Reference Manual

Granular synthesizer / sampler / looper / delay for Pure Data

Steven Benjamin

OVERVIEW

ligase~ is a granular synthesizer/sampler/looper/delay with real-time recording, reel/splice management, spectral fog, filter, distortion and chaotic parameter modulation inspired by features of the Morphagene, Audiomulch's DLGranulator and Tidalcycles.

ligase~ implements asynchronous granular synthesis with splice-based sample organization. The engine operates on a fixed-size stereo buffer (10 minutes at the host sample rate) divided into segments called splices.

Components

Reel: Circular stereo buffer (MAX_REEL_SECONDS * 48000 samples)

Scheduler: Grain pool manager (configurable 1-2000 grains, default 200)

Recorder: Real-time input capture with sound-on-sound mixing

Envelope: Table-based amplitude shaping (parabolic, trapezoidal, cosine)

Grain pool allocation is configurable via ligase.conf file placed in the same directory as the external.

Format: max_grains = N (range 1-2000).

Grain Generation

1. Trigger timing controlled by IOT (interonset time): 0.0005-2.0 seconds
2. Grain size: 0.001-2.0 seconds (default 0.1)
3. Playhead position determines grain start point within current splice
4. Speed parameter controls playback rate and direction (-4.0 to 4.0)
5. Envelope applied via table lookup with linear interpolation
6. Maximum concurrent grains: 1 to pool_size

Grain Mixing

Constant-power stereo panning (0=left, 0.5=center, 1=right)

Per-grain amplitude scaling: 0.0-2.0 (default 1.0)

Summation of all active grains per sample

No automatic gain compensation

INLETS

1. Audio L - Left input + Messages + Bangs(metro/clock)
2. Audio R - Right input
3. GrainSize - Grain duration (0.001-10 seconds)
4. GrainStart - Normalized start position (0-1)
5. Speed - Playback rate (-4 to 4)
6. Organize - Splice selection (0-1)
7. ScanRate - Scan speed for mode 2 (-1000 to 1000)
8. SoS - Sound-on-sound mix (0-1)
9. IOT - Interonset time (0.0005-2 seconds)
10. MaxGrains - Max concurrent grains (1-pool_size)
11. GDelay Time - Delay time (0-10 seconds)
12. GDelay Feedback - Feedback amount (0-1) (DD-4/Bencina only)
13. GDelay Tone - Lowpass filter (0-1) (DD-4/Bencina only)
14. GDelay Mix - Dry/wet mix (0-1)
15. Fog Mix - Spectral fog blend (0-1)
16. Moog Cutoff - Filter cutoff (20-20000 Hz)
17. Moog Resonance - Filter resonance (0-4)
18. Moog Mix - Filter dry/wet (0-1)
19. MIDI Pitch - MIDI note (1-127, 0=inactive)
20. Envelope Skew - Shape skew (0-1)
21. Amplitude - Grain amplitude (0-2)
22. Pan - Stereo position (0-1)

Unconnected inlets default to stored values (set via messages).

Zero values from unconnected inlets do not overwrite stored parameters.

OUTLETS

1. Signal - Audio output left
2. Signal - Audio output right
3. Bang - Splice end/wrap notification (modes 2 & 3)
4. Bang - Grain onset (every N grains, set via grain_bang_rate)

MESSAGES

File Operations

load - Load a 32-bit float stereo WAV (any rate; plays at the host rate). Resolved relative to the patch directory. save - Save current reel to WAV

Playback Control

play <0|1> - Start/stop playback record <0|1> - Start/stop recording recsplice - Record and create splice on stop recinput - Input-only recording, create splice on stop

Recording Configuration

sos <0-1> - Sound-on-sound mix level sos_mode <0|1> - 0=record only, 1=crossfade (default)

Splice Navigation

shift - Move splice selection (integer) organize <0-1> - Select splice by normalized value splice - Add splice marker at current position

Splice Management

clear_splices - Remove all markers, clear buffer clear_splices_except_current - Keep only current splice clear_current_splice - Remove current splice splice_join_right - Join current with right neighbor splice_join_all - Join all splices into one

Splice Behavior

splice_create_pos <0|1|2> - 0=playback pos, 1=right, 2=end splice_jump <0|1> - Jump to new splice on creation splice_finish_nav <0|1> - Finish playback before nav splice_split <0|1> - Allow/prevent splitting current

Playhead

playhead <1|2|3> - Static/Scanning/Clock Advance scanrate - Scan rate for mode 2 speed <-4 to 4> - Playback speed/direction clock_advance_quant <0|1> - Use current/quantized length clockstop - Stop clock, keep last BPM

Grain Parameters

grainsize <0.001-10> - Grain duration in seconds grainstart <0-1> - Position within splice where grains begin (0=start, 1=end) iot <0.0005-2> - Interonset time in seconds maxgrains <1-pool_size> - Max concurrent grains amplitude <0-2> - Grain amplitude pan <0-1> - Stereo position (0=left, 0.5=center, 1=right) pan_mode <0|1> - Mono panning / Stereo balance saw_cycles <0-64> - Saw envelope modulation cycles saw_depth <0-1> - Saw envelope modulation depth envelope <0|1|2|3|4> - Parabolic/Trapezoidal/Cosine/Gaussian/Exponential env_skew <0-1> - Envelope shape skew grain_bang_rate - Bang outlet every N grains (0=off)

Envelope

envelope <0|1|2> - Parabolic/Trapezoidal/Cosine env_skew <0-1> - Envelope shape skew

IOT Quantization

timesig / - Time signature quantize <1|2|4|8|16|32|64|128> - Note subdivision quant <0-1> - Quantization amount

Grain Size Quantization

gs_timesig / gs_quantize gs_quant <0-1>

Delay Quantization

delay_timesig / delay_quantize delay_quant <0-1>

Grain Delay

gdelay_time <0-9.5> - Delay time in seconds gdelay_feed <0-1> - Feedback amount gdelay_tone <0-1> - Lowpass filter (0=dark, 1=bright) gdelay_mix <0-1> - Dry/wet mix gdelay_clear - Clear delay buffer

Distortion

distortion <0-1> - Drive intensity distortion_enable <0|1> - Enable/bypass distortion_oversampling <1|2|4|8> - Oversampling factor distortion_position <0|1> - Per-grain/Post-mix dist_waveshaper_mode <0|1|2|3|4> - Tanh/Arctan/Asymmetric/Blend/Polynomial distortion_pre_hp_freq <30-500> - Pre-HP frequency distortion_pre_hp_mix <0-1> - Pre-HP mix distortion_post_lp_freq <2400-10000> - Post-LP frequency distortion_post_lp_mix <0-1> - Post-LP mix distortion_notch_freq - Notch center frequency distortion_notch_bw - Notch bandwidth distortion_notch_mix <0-1> - Notch mix dist_emphasis_mode <0|1> - HP/LP pre-emphasis dist_emphasis_freq <100-5000> - Emphasis frequency dist_pregain <0.1-10> - Pre-saturation gain dist_curve_blend <0-1> - Tanh/arctan morph dist_drive_pos <1-20> - Positive drive (asymmetric) dist_drive_neg <1-20> - Negative drive (asymmetric) dist_poly_c1 <-10 to 10> - Linear coefficient dist_poly_c2 <-10 to 10> - Quadratic coefficient dist_poly_c3 <-10 to 10> - Cubic coefficient

Fog (Spectral Effect)

fog_smear_bins <0-32> - Frequency smear width (bins) fog_smear_enable <0|1> - Enable/disable smear fog_smear_onset_curve <0|1|2> - Linear/Exponential/Logarithmic fog_smear_onset_amount <0-1> - Smear onset depth fog_mag_cutoff <0.1-20> - Magnitude filter cutoff (Hz) fog_mag_resonance <0.1-10> - Magnitude filter Q fog_phase_cutoff <0.1-20> - Phase filter cutoff (Hz) fog_specmagfilter_enable <0|1> - Enable/disable temporal filter fog_specmagfilter_onset_curve <0|1|2>

- Linear/Exponential/Logarithmic fog_specmagfilter_onset_amount <0-1> - Temporal filter onset depth fog_stereo_filter_mode <0|1> - Shared/Independent stereo state fog_position <0|1> - Per-grain/Post-mix mode

Moogladder Filter

moog_cutoff <20-20000> - Cutoff frequency (Hz) moog_resonance <0-4> - Resonance amount moog_mix <0-1> - Dry/wet mix moog_enable <0|1> - Enable/bypass

Delay Modes

delay_mode <0|1|2> - DD-4/Bencina/Stut stut - Trigger stut effect stut_reps <1-16> - Stut repetitions stut_reduction <0-1> - Gain decay per repeat stut_spacing - Spacing between repeats bencina_iot - Bencina grain spacing bencina_grainsize - Bencina grain size bencina_wrap <0|1> - Global/Splice wrap mode bencina_clear - Clear bencina grains

Sphere Simulation

sphere_kick - Apply velocity impulse sphere_damping <0-1> - Damping coefficient sphere_elasticity <0-1> - Bounce elasticity sphere_reset - Reset to initial state sphere_mode <0-6> - Output mode (X/Y/Z/VelX/VelY/VelZ/VelMag)

Parameter Ranging

param_range - Set parameter range rand_type - Assign generator param_lock … - Disable modulation and lock parameter(s) at their current modulated value. param_invert <0|1> - Invert a parameter's modulation output (mirror around its range). param_slew <0-1> - Exponential smoothing on a parameter's modulation output. param_base_value - Set the PERLIN_2D Y-coordinate (decorrelation) for a parameter.

Generators

rand_1, rand_2, rand_3, rand_4 perlin_1d_1, perlin_1d_2, perlin_1d_3, perlin_1d_4 perlin_2d_1, perlin_2d_2, perlin_2d_3, perlin_2d_4 lorenz_1, lorenz_2, lorenz_3, lorenz_4 nbody_1, nbody_2, nbody_3, nbody_4 sphere_1, sphere_2, sphere_3, sphere_4 saw_1, saw_2, saw_3, saw_4 sine_1, sine_2, sine_3, sine_4 square_1, square_2, square_3, square_4

Generator Modulation Outlet Send

The four send outlets (modout1–modout4) are configured like any modulatable parameter — by addressing the param name modoutN with the unified messages. There are no dedicated modout_source/modout_range commands: rand_type modoutN - assign a generator to outlet N param_range modoutN - set the outlet's output range

Noise Frequency

noise_freq - Set all generator speeds noise_freq_1|2|3|4 - generator instance speeds

N-Body Configuration

nbody_mode - Set output mode (0-10) nbody_epsilon <0.01-1> - Softening parameter nbody_damping <0-0.1> - Energy damping nbody_pump <0-0.01> - Energy injection nbody_reset - Reset to initial conditions

Sphere Configuration

sphere_kick - Apply velocity impulse sphere_damping <0-1> - Damping coefficient sphere_elasticity <0-1> - Bounce elasticity sphere_reset - Reset to initial state sphere_mode <0-6> - Output: X/Y/Z/VelX/VelY/VelZ/VelMag

Pitch Control

pitch_mode <0|1|2|3|4> - Off/Semitones/Range/Scale/MIDI pitch_semitones <-24 to 24> - Fixed transposition pitch_range - Semitone range pitch_rand_type - Random generator for pitch pitch_scale … - Scale definition (semitones)

Defaults

Core Parameters

grainsize: 0.1 seconds grainstart: 0.5 (center of splice) speed: 1.0 (normal forward) iot: 0.1 seconds maxgrains: 4 amplitude: 1.0 pan: 0.5 (center) pan_mode: 0 (constant-power mono panning) saw_cycles: 0.0 (off) saw_depth: 0.0 (off) env_skew: 0.5 (symmetric)

Envelope

type: ENVELOPE_COSINE table_length: 4096 samples

Playhead

mode: PLAYHEAD_MODE_STATIC (mode 1) scanrate: 1.0

Recording

sos: 0.5 (50/50 mix) sos_mode: 1 (Morphagene-style crossfade) mode: RECORD_MODE_OVERDUB

Splice

create_pos: 0 (at playback position) jump: 0 (stay in current) finish_nav: 0 (immediate) split: 0 (allow split)

Effects

fog_mix: 0.0 (dry) fog_smear_bins: 8 fog_smear_enable: 1 fog_smear_onset_curve: 2 (logarithmic) fog_smear_onset_amount: 1.0 fog_mag_cutoff: 2.0 Hz fog_mag_resonance: 1.0 fog_phase_cutoff: 2.0 Hz fog_specmagfilter_enable: 1 fog_specmagfilter_onset_curve: 2 (logarithmic) fog_specmagfilter_onset_amount: 1.0 fog_stereo_filter_mode: 0 (shared) fog_position: 1 (post-mix) fog_pool_size: 4 (configurable via ligase.conf) gdelay_time: 0.0 (off) gdelay_feed: 0.0 gdelay_tone: 0.5 gdelay_mix: 0.0 delay_mode: 0 (DD-4) delay_quant_note: 16 (1/16) delay_quant_amount: 0.0 (disabled) stut_reps: 4 stut_reduction: 0.5 stut_spacing: 62.5 ms bencina_iot: 50.0 ms bencina_grainsize: 0.1 seconds bencina_wrap: 1 (splice) distortion: 0.0 (clean) distortion_enable: 1 distortion_oversampling: 4x distortion_position: 1 (post-mix) moog_cutoff: ~1000 Hz moog_resonance: 0.0 moog_mix: 0.0 moog_enable: 1

Distortion Filters

pre_hp_freq: 30 Hz pre_hp_mix: 0.5 post_lp_freq: 16000 Hz post_lp_mix: 0.5 notch_freq: 3000 Hz notch_bandwidth: 500 Hz notch_mix: 0.0 (inactive) emphasis_freq: 800 Hz emphasis_mode: 0, EMPHASIS_MODE_HP pregain: 1.0

Pitch Control

mode: PITCH_MODE_OFF semitones: 0.0 midi_note: 60 (C4)

Quantization

All quant_amount: 0.0 (off) timesig: 4/4 subdivision: 4 (quarter note)

Noise Generators

All frequency_scale: 1.0 (normal) All rand_type: RAND_TYPE_RAND All rand_instance: 0 (generator 1)

Pool

pool_size: 200 grains (configurable via ligase.conf)

Buffer

Sample rate: follows the host (Pd) rate — the reel capacity and saved WAVs use it (44.1/48/96 kHz etc.) Max length: 600 seconds Channels: 2 (stereo)

PLAYBACK CONTROL

Transport

play Start/stop grain triggering. 0 stops, non-zero starts.

When started:

- Checks reel length > 0. Posts error and aborts if empty.
- Initializes playback_position to current splice start.
- Sets is_playing and is_triggering flags.

When stopped:

- Clears is_playing and is_triggering.
- Active grains finish their envelopes.

Default: 0 (stopped)

record Overdub into the CURRENT splice — Time Lag Accumulation. Argument: 0 stops, non-zero starts.

The record head loops within the current splice bounds, re-recording what is heard (the granular playback of the splice, mixed with the live input) back into the same splice. Because the splice being recorded is also the one being granulated, successive passes accumulate — layers build and any pitch transposition compounds across loops. SOS sets the input-vs-feedback balance (the feedback amount); the feedback is held just below unity and soft-limited so it sustains/decays rather than running away. Cross-splice recording is done by shifting the current splice (which moves the bounds) — overdub never extends the reel.

- Sets RECORD_MODE_OVERDUB; record head wraps within the current splice
- Stop: posts “recording stopped” (no new splice created)

recsplice Record a NEW splice of what is heard.

- Sets RECORD_MODE_NEW_SPLICE
- Records the SOS-mixed monitor signal (live input crossfaded with the granular playback, per SOS) — i.e. “what you hear”. SOS acts as a VCA on the recorded signal.
- Posts starting position (current reel length); creates the splice when recording stops

recinput Record the live INPUT only — the one mode where SOS does NOT act as a VCA.

- Sets RECORD_MODE_INPUT_ONLY

- Captures the raw input directly (no SOS, no feedback); ignores the sos parameter
- Creates a splice when stopped

Playhead Modes

playhead 1|2|3 Set playhead behavior.

Mode 1: Static (default)

Playback position fixed. grain_start parameter (inlet 4) sets grain spawn location directly Grain position: splice_start + (grain_start × splice_length)

Detects grain_start parameter wraps (change > 0.5), bangs outlet 3. No playback_position advancement.

Mode 2: Scanning

Playback position advances continuously at scan_rate (samples per sample).

Grain position: playback_position + (grain_start × splice_length)

Advancement: playback_position += scan_rate every sample

Wraps at splice boundaries, bangs outlet 3 on wrap. Checks pending splice navigation after wrap.

Negative scan_rate = reverse.

Mode 3: Clock Advance

Playback position advances only when clock bang received (inlet 1).

Advance amount: grain length × sample_rate

Grain length source: current or quantized (see clock_advance_quant)

Grain position: playback_position + (grain_start × splice_length) between bangs Wraps, bangs outlet, checks pending navigation like Mode 2.

scanrate Set scanning speed for Mode 2.

1.0 = real-time forward

-1.0 = real-time reverse

0.0 = static

Default: 1.0

clock_advance_quant 0|1 Select grain length sourcefor Mode 3 advance.

0 = current actual grain length (varies per grain)

1 = quantized grid length (requires valid BPM and gs_quant_grid_ms > 0)

Default: 0

Splice Structure

- Array of position markers (0-300 maximum)
- Each splice defined by start marker, extends to next marker or buffer end
- Current splice selected via organize parameter (0.0-1.0 maps to splice index)
- Shift message: relative navigation (shift 1 = next, shift -1 = previous)

Splice Bounds

Grains read exclusively within current splice boundaries. Position wraps at splice_end back to splice_start. This creates independent loop regions within the buffer.

Splice Navigation

organize Navigate to splice by normalizedposition (0.0-1.0).

Maps to index: int(value × (splice_count - 1))

Clamps input to [0.0, 1.0]

If splice_finish_nav = 0: immediate navigation, resets playback_position to splice start

If splice_finish_nav = 1: queues navigation, executes after current splice wraps

shift Move splice selection by delta.

Adds delta to current_splice index.

Clamps to [0, count-1]. No wraparound.

Examples: shift 1 = next, shift -1 = previous

If splice_finish_nav = 1: queues as pending

If splice_finish_nav = 0: immediate

Splice Creation

splice Create splice marker.

Position determined by splice_create_pos:

- 0: at playback_position (if 0 and reel exists, uses reel_length)

- 1: at current splice end boundary
- 2: at reel end

Rate limited: minimum 10ms between creation

Posts error if MAX_SPLICES (64) reached

splice_create_pos 0|1|2 Set splice creation positionmode.

0 = at playback position (default)

1 = right of current splice boundary

2 = at end of reel (Morphagene append)

splice_jump 0|1 Control navigation after creatingnew splice.

0 = stay in current splice (default)

1 = jump to newly created splice immediately

splice_finish_nav 0|1 Control navigation timing fororganize and shift messages.

0 = immediate navigation (default)

1 = queue navigation, execute after current splice wraps

splice_split 0|1 Control splice splitting behavior.

0 = allow splitting current splice (default)

1 = preserve current splice length

RECORDING CONFIGURATION

Record Modes

Three recording modes, set by message commands. Mode determines recording behavior and splice creation.

record Overdub mode. Argument: 0 stops, non-zerostarts.

- Sets RECORD_MODE_OVERDUB
- Start: positions record_position at current_splice_start
- Records into current splice bounds
- Continues past splice boundary into adjacent splice (cross-splice recording)
- Mixing controlled by crossfade_mix (sos value)
- Stop: Posts “recording stopped” only

recsplice New splice mode.

- Sets RECORD_MODE_NEW_SPLICE
- Start: positions record_position at reel length (append)
- Stores new_splice_start = current reel length
- Mixing controlled by crossfade_mix (sos value)
- Stop: Creates splice marker at new_splice_start position, sets as current splice
- Posts splice creation confirmation with position

recinput Input-only mode.

- Sets RECORD_MODE_INPUT_ONLY
- Start: positions record_position at reel length (append)
- Stores new_splice_start = current reel length
- Records input directly, ignores sos/crossfade_mix completely
- Stop: Creates splice marker at new_splice_start position, sets as current splice

- Posts splice creation confirmation with position

Default mode: RECORD_MODE_OVERDUB

SOS Mode Control

sos_mode 0|1 Set Sound-on-Sound operational mode.

Mode 0: Record Only

- Output: Constant-power crossfade between input and granular
  - sos = 0.0: 100% granular (input_gain = 0.0, granular_gain = 1.0)
  - sos = 0.5: Equal power mix (both ≈ 0.707)
  - sos = 1.0: 100% input (input_gain = 1.0, granular_gain = 0.0)
  - Formula: out = input × sin(sos × π/2) + granular × cos(sos × π/2)
- Recording: happens before output mixing
  - Records raw input only (not the mixed output)
  - crossfade_mix forced to 1.0 (full replacement, no sound-on-sound)
  - sos parameter does NOT affect recording in this mode
  - Always replaces buffer content with input signal
- Output and recording are decoupled:
  - Recording: Raw input → buffer (replace mode)
  - Output: Input + granular constant-power mix (monitoring)

Mode 1: Morphagene (default)

- Output: Constant-power crossfade between input and granular
  - output = input × sin(sos × π/2) + granular × cos(sos × π/2)
  - sos = 0.0: 100% granular (input_gain = 0.0, granular_gain = 1.0)
  - sos = 0.5: Equal power mix (both ≈ 0.707)
  - sos = 1.0: 100% input (input_gain = 1.0, granular_gain = 0.0)
  - sos from signal inlet (inlet 8) or stored sos_value
  - Uses inlet value if non-zero, else uses stored value

  - Clamped to [0.0, 1.0]
  - NaN/Inf protection via isfinite() check
- Recording: captures output signal
  - What you hear is what gets recorded
  - Forces crossfade_mix = 1.0 during recording (replace mode)
  - Restores original crossfade_mix after recording
- When no playback (reel length = 0):
  - Output = input × sos (linear VCA behavior)
  - Records scaled input

Default: 1 (Morphagene mode)

SOS Crossfade Parameter

sos Set crossfade mix value.

Range:

0.0 to 1.0 (clamped in processing) Default: 0.5

SPLICE NAVIGATION

Direct Navigation

shift Navigate by relative offset.

Adds delta to current_splice index. Clamps to [0, count-1]. No wraparound at boundaries.

- Positive: move right through array
- Negative: move left through array
- 0 stops at first splice
- count-1 stops at last splice

If finish_before_nav = 1: calculates target with wraparound ((target % count) + count) % count, stores as pending, executes after playback wraps at splice boundary.

If finish_before_nav = 0: immediate navigation.

organize Navigate by normalized position.

Maps input [0.0-1.0] to discrete splice index: int(value × (count - 1))

Clamps input to [0.0, 1.0] before mapping.

Examples: 0.0 = first splice, 0.5 = middle splice, 1.0 = last splice

If finish_before_nav = 1: calculates target, stores as pending.

If finish_before_nav = 0: immediate navigation.

Splice Creation

splice Create splice marker.

Position determined by create_position setting:

Mode 0 (default): At playback_position - If playback_position = 0 and reel exists, uses reel_length instead - If split_current = 1 and position would split current splice (between splice_start and

splice_end), moves to splice_end

Mode 1: At current splice end boundary (splice_end) Mode 2: At reel end (reel_length) Rate limited: minimum 10ms between creation. Posts error if violated.

Maximum: 64 splices (MAX_SPLICES). Posts error if exceeded.

After creation, if jump_to_new = 1: sets current_splice to new index, resets playback_position to new splice start.

splice_create_pos 0|1|2 Set creation position mode. 0 = at playback position

1 = right of current splice

2 = at end of reel

Default: 0

splice_jump 0|1 Control navigation after splice creation.

0 = stay in current splice (default)

1 = jump to newly created splice, reset playback_position

splice_finish_nav 0|1 Control navigation timing forshift and organize.

0 = immediate navigation (default)

1 = queue navigation, execute when playback wraps at splice boundary

Pending navigation stored in pending_splice. Cleared after execution.

splice_split 0|1 Control current splice splitting.

0 = allow splitting current splice (default)

1 = preserve current splice length

When enabled (mode 1), if create_position = 0 and calculated position falls within current splice bounds, position moves to splice_end.

Splice Removal

clear_splices Remove all splice markers, clear entire audio buffer.

Calls splice_clear() and reel_clear(). Destructive operation. Sets count = 0, current_splice = 0.

clear_splices_except_current Remove all splices except current, clear buffer

except current splice audio.

Destructive. Extracts current splice bounds, clears buffer outside bounds, reorganizes markers:

- If current has right neighbor: creates two markers (position 0 and splice_length)
- If current extends to reel end: creates one marker (position 0)

Current splice moves to position 0. Sets current_splice = 0.

clear_current_splice Remove current splice marker immediately.

Uses splice_remove() to delete marker at current_splice index. Shifts subsequent markers down. Active grains finish naturally.

Adjusts current_splice index if needed:

- If count becomes 0: current_splice = 0
- If current_splice >= count: current_splice = count - 1

Posts count of remaining splices.

Splice Joining

splice_join_right Join current splice with right neighbor.

Calculates right neighbor with wraparound: (current_splice + 1) % count

Removes marker at right neighbor (boundary between them). Updates current_splice index if right_splice <= current_splice.

No effect if only one splice exists.

splice_join_all Join all splices into one continuoussplice. Calls splice_clear(). Entire reel becomes single splice with no markers. Sets count = 0, current_splice = 0.

Posts number of splices joined.

PLAYHEAD MODES

Three modes control how playback position advances and where grains spawn within the current splice.

playhead 1|2|3 Set playhead mode.

Mode 1: Static (default)

Playback position fixed. grain_start signal (inlet 4) directly controls grain spawn location.

Grain position: splice_start + (grain_start × splice_length)

grain_start range: 0.0 to 1.0

0.0 = splice start, 0.5 = splice middle, 1.0 = splice end

Wrap detection: When grain_start change exceeds 0.5 (indicating wrap from 1.0→0.0 or 0.0→1.0), bangs outlet 3.

playback_position: Never advances. Remains at value set by play message (splice start).

grain_start tracking: Stores prev_grain_start each block for wrap detection.

Mode 2: Scanning

Playback position advances continuously. grain_start offsets grain spawn relative to moving playhead.

Grain position: playback_position + (grain_start × splice_length) Wraps to stay within splice bounds.

Advancement: playback_position += scan_rate every audio sample

scan_rate units: samples per sample

1.0 = real-time forward

1.0- = real-time reverse

0.0 = static (no advancement)

2.0 = double-speed forward

Boundary wrapping:

When playback_position >= splice_end: wraps by subtracting splice_length

When playback_position < splice_start: wraps by adding splice_length

Bangs outlet 3 on wrap. Checks pending_splice navigation after wrap, executes if queued.

Mode 3: Clock Advance

Playback position advances only when clock bang received (inlet 1). Grains spawn at static position between bangs.

Grain position: playback_position + (grain_start × splice_length) Wraps to stay within splice bounds.

Advancement trigger: Bang to inlet 1 sets clock_bang_received flag.

Advance amount: grain_length × sample_rate (converted from seconds to samples)

Grain length source:

Determined by clock_advance_quant setting and current state.

If clock_advance_quant = 0: uses quantized_grain_size (current grain length from perform cycle)

If clock_advance_quant = 1 AND gs_quant_grid_ms > 0 AND bpm > 1.0: Uses gs_quant_grid_ms / 1000.0 (quantized grid length in seconds)

Otherwise: uses quantized_grain_size

Per-bang behavior:

Flag checked once per perform cycle. If set, advances playback_position, wraps at boundaries, bangs outlet 3 if wrapped, checks pending navigation, then clears flag.

Grains trigger normally at IOT rate at current playback_position while flag is clear.

Control Messages

scanrate Set scan rate for Mode 2.

Applied per audio sample as: playback_position += scan_rate

Default: 1.0

clock_advance_quant 0|1 Select grain length sourcefor Mode 3 advancement.

0 = use current grain length (varies per grain)

1 = use quantized grid length (requires valid BPM and gs_quant_grid_ms)

Default: 0

GRAIN PARAMETERS

Interonset Time

iot Set time interval between grain starts.

Range: 0.0005 to 2.0 seconds

Clamped to bounds.

Converted to grain_trigger_period: int(iot × sample_rate) samples

When iot < grain_size: grains overlap

When iot > grain_size: grains separate

When iot = grain_size: continuous stream

Signal inlet 8: updates if value in range [0.001, 2.0], else keeps current value.

Quantization: If quant_amount > 0 and BPM valid:

grain_trigger_period = base_period × (1 - quant_amount) + quant_period × quant_amount

Minimum grain_trigger_period enforced: 1 sample (prevents CPU overload)

Default: 0.1 seconds (100ms)

Grain Size

grainsize Set grain duration.

Range: 0.001 to 10.0 seconds

Clamped to bounds.

Signal inlet 2: updates if value in range [0.001, 10.0], else keeps current value.

Default: 0.1 seconds (100ms)

Grain Start Position

grainstart Set normalized grain spawn positionwithin splice.

Range: 0.0 to 1.0

0.0 = splice start, 0.5 = splice middle, 1.0 = splice end

Signal inlet 4: updates if value in range [0.0, 1.0]

Special handling: if inlet reads 0.0 and stored value is non-zero, assumes inlet unconnected and preserves stored value.

Interacts with playhead mode:

- Mode 1 (Static): grain_pos = splice_start + (grainstart × splice_length)
- Mode 2 (Scanning): grain_pos = playback_position + (grainstart × splice_length)
- Mode 3 (Clock): grain_pos = playback_position + (grainstart × splice_length)

Default: 0.5 (middle)

Maximum Grains

maxgrains Set maximum concurrent active grains. Range: 1 to pool_size

pool_size read from ligase.conf (default: 200)

Clamped to [1, pool_size]

Controls polyphony limit. Grain scheduler uses free-list allocation. When limit reached, oldest grains replaced.

Signal inlet 9: updates if value in range [1, pool_size], else keeps current value.

Default: 4

Amplitude

amplitude Set grain output level.

Range: 0.0 to 2.0

Clamped to bounds.

Applied during grain triggering. Passed to scheduler_trigger_grain().

Signal inlet 19: amplitude_in (read from perform cycle)

Default: 0.75

Per-grain amplitude scaling and randomization

There are two operational modes based on param_range configuration:

Mode 1: Direct Control (param_range disabled, default)

- Uses amplitude value directly from inlet or message

- No per-grain randomization
- Consistent amplitude across all grains

Mode 2: Modulated Range (param_range enabled)

- Samples amplitude from [min, max] range per grain
- Uses inlet/message value as base for sampling
- Random source determines variation pattern
- Examples:

param_range amplitude 0.5 1.0 → grains vary 0.5-1.0

rand_type perlin_1d_1 amplitude → smooth amplitude evolution

rand_type rand_1 amplitude → random per-grain levels

Pan Mode

pan_mode Select panning algorithm.

Mode 0: Constant-Power Mono Panning (default)

Sums stereo source to mono before panning

Each grain becomes a point source in the stereo field

Ideal for spatial positioning and grain cloud diffusion

Mode 1: Stereo Balance

Preserves stereo width of source material

Applies constant-power gains independently to L/R channels

Maintains stereo image character while adjusting balance

Ideal for stereo sources where width should be preserved

Default: 0 (constant-power mono panning)

Pan Law

Both modes use constant-power panning to maintain consistent perceived loudness:

pan_angle = pan * (π/2)

left_gain = cos(pan_angle) // 1.0 → 0.707 → 0.0

right_gain = sin(pan_angle) // 0.0 → 0.707 → 1.0

This ensures $\cos^2(\theta) + \sin^2(\theta) = 1$ at all pan positions, preventing the "hole in the middle" effect where center-panned signals sound quieter.

At center (pan=0.5): Both channels at 0.707 (-3dB) = 0dB total power

Pan Modulation

Pan can be modulated per-grain using param_range:

Direct Control (param_range disabled)

Uses pan value directly from inlet or message

No per-grain randomization

All grains at same stereo position

Default behavior

Modulated Range (param_range enabled):

Samples pan position from [min, max] range per grain

Uses inlet/message value as base for sampling

Random source determines spatial pattern

Creates spatial diffusion effects

Examples:

param_range pan 0.0 1.0

rand_type perlin_1d_1 pan ← smooth stereo movement

param_range pan 0.3 0.7

rand_type rand_1 pan ← random center-focused placement

param_range pan 0.0 0.5

rand_type lorenz_1 pan ← chaotic left-side positioning

Spatial Cloud Effects:

Wide range + fast modulation = diffuse stereo cloud

Narrow range + slow modulation = subtle stereo shimmer

Per-grain randomization creates spatial depth impossible with traditional panning

Envelope Shape

envelope 0|1|2 Set grain envelope type.

Type 0: Parabolic (Welch window)

Formula: $1 - 4(x - 0.5)^2$

Smooth, gentle envelope. Quadratic curve.

Type 1: Trapezoidal

Linear attack (10% default), sustain, linear release (10% default). Hard attack/release with flat sustain. Percentages adjusted by skew.

Type 2: Cosine (Hann window, default)

Formula: $0.5 \times (1 - \cos(2\pi \times x))$

Smoothest envelope. Reduces spectral artifacts.

Envelope table length: 4096 samples

Preserves skew value when type changes.

Default: ENVELOPE_COSINE (type 2)

Envelope Skew

Skew remaps time axis before envelope calculation, shifting peak position while maintaining smoothness.

env_skew Set envelope attack/decay balance.

Range: 0.0 to 1.0

Clamped to bounds.

0.5 = symmetric (peak at center)

< 0.5 = shorter attack, longer decay (peak shifts left)

0.5 = longer attack, shorter decay (peak shifts right)

Regenerates envelope table immediately on change.

Signal inlet 18: updates if value in range [0.001, 1.0] and different from current, else keeps current value.

Saw Envelope Modulation

Per-grain sawtooth wave carved into the envelope. Creates jagged, rhythmic amplitude spikes within each grain. The saw wave modulates the envelope shape: at depth=0 the envelope is unmodified, at depth=1 the saw wave fully carves the envelope into sharp peaks.

Formula: env_val = base_env * (1.0 - depth * (1.0 - saw_val))

saw_cycles Number of sawtooth cycles per grain duration.

Range: 0.0 to 64.0. Default: 0.0 (off).

0.0 = no modulation. 1.0 = one ramp per grain. Higher values = more spikes. Non-integer values produce partial final cycles.

Signal inlet 21 or message. Supports param_range modulation.

saw_depth Modulation intensity.

Range: 0.0 to 1.0. Default: 0.0 (off).

0.0 = pure base envelope. 1.0 = maximum jaggedness (spikes drop to zero between peaks).

Signal inlet 22 or message. Supports param_range modulation.

Grain Bang Output

grain_bang_rate Set grain onset bang rate.

Range: 0 to 1000

0 = disabled (no bangs)

1 = bang every grain

2 = bang every 2nd grain

N = bang every Nth grain

Bangs outlet 4 when grain_bang_counter reaches grain_bang_rate. Counter resets on rate change.

TIMING & QUANTIZATION

Clock Management

Clock bangs to inlet 1 establish tempo. BPM calculated from the interval between bangs.

Bang to inlet 1: Calculate BPM and update quantization grids.

First bang: stores timestamp, no BPM calculation

Second bang onward: calculates interval, derives BPM

BPM calculation: 60000.0 / interval_ms

Sets clock_running flag. Recalculates all active quantization grids (IOT, grain size, delay) using new BPM.

In playhead Mode 3: sets clock_bang_received flag for playhead advance.

clockstop Clears clock_running flag.

Preserves last calculated BPM and quantization grids for continued use. Posts current BPM value.

IOT Quantization

Snaps interonset time to rhythmic grid.

timesig Set time signature for IOT quantization.

Format: “numerator/denominator” (e.g., “7/5”, “4/4”) Requires symbol argument. Parses using sscanf().

Validation: numerator > 0, denominator > 0

Posts error if invalid format or values.

Stores: time_sig_numerator, time_sig_denominator

Default: 4/4

quantize Set IOT quantization note subdivision. Valid values: 1, 2, 4, 8, 16, 32, 64, 128

Represents note division (16 = 1/16 note)

Stores quant_note. Recalculates quant_grid_ms if BPM > 0:

Grid calculation:

ms_per_whole_note = (60000.0 / bpm) × 4.0

quant_grid_ms = ms_per_whole_note / quant_note

Posts grid size in milliseconds if BPM available, otherwise defers calculation.

Default: 16 (1/16 note)

quant Set IOT quantization amount (blend).

Range: 0.0 to 1.0

0.0 = no quantization (free timing)

1.0 = full quantization (locked to grid)

0.5 = 50% blend

Applied in perform cycle:

grain_trigger_period = base_period × (1 - quant_amount) + quant_period × quant_amount

Requires: quant_amount > 0, bpm > 1.0, quant_grid_ms > 0

Default: 0.0 (no quantization)

Grain Size Quantization

Snaps grain duration to rhythmic grid.

gs_timesig Set time signature for grain size quantization.

Format: “numerator/denominator”

Validation and parsing identical to timesig.

Stores: gs_time_sig_numerator, gs_time_sig_denominator

Default: 4/4

gs_quantize Set grain size quantization notesubdivision. Valid values: 1, 2, 4, 8, 16, 32, 64, 128

Calculation identical to quantize.

Stores gs_quant_note. Recalculates gs_quant_grid_ms if BPM > 0.

Default: 16 (1/16 note)

gs_quant Set grain size quantization amount.

Range: 0.0 to 1.0

Blend between free grain_size and quantized grid.

Applied in perform cycle:

quant_grain_size_sec = gs_quant_grid_ms / 1000.0

quantized_grain_size = grain_size × (1 - gs_quant_amount) + quant_grain_size_sec × gs_quant_amount

Result assigned to scheduler->grain_size.

Requires: gs_quant_amount > 0, bpm > 1.0, gs_quant_grid_ms > 0

Default: 0.0 (no quantization)

Delay Time Quantization

Snaps grain delay time to rhythmic grid.

delay_timesig Set time signature for delayquantization.

Format: “numerator/denominator”

Validation and parsing identical to timesig.

Stores: delay_time_sig_numerator, delay_time_sig_denominator

Default: 4/4

delay_quantize Set delay quantization note subdivision. Valid values: 1, 2, 4, 8, 16, 32, 64, 128

Calculation identical to quantize.

Stores delay_quant_note. Recalculates delay_quant_grid_ms if BPM > 0.

Default: 16 (1/16 note)

delay_quant Set delay quantization amount.

Range: 0.0 to 1.0

Blend between free delay_time and quantized grid.

Applied in perform cycle:

quant_delay_sec = delay_quant_grid_ms / 1000.0

quantized_delay = current_delay × (1 - delay_quant_amount) + quant_delay_sec × delay_quant_amount

Result assigned to grain_delay->delay_time.

Requires: delay_quant_amount > 0, bpm > 1.0, delay_quant_grid_ms > 0

Default: 0.0 (no quantization)

PITCH & SPEED

pitch_mode 0|1|2|3|4 Five pitch modes control relationship between speed parameter and playback rate.

Mode 0 - OFF (default)

speed

Speed parameter = playback rate directly.

speed_range active if enabled.

Formula: final_speed = sample_param_range(speed_range, speed_inlet)

Mode 1 - SEMITONES

Speed parameter = base pitch.

Fixed semitone transposition applied.

speed_range bypassed.

Formula: final_speed = speed * 2^(semitones/12)

pitch_semitones <-24 to 24>

Example: pitch_semitones 7 = perfect fifth up

Mode 2 - RANGE

Speed parameter = base pitch.

Random semitone shift from range.

speed_range bypassed.

Formula: final_speed = speed * 2^(random_semitones/12)

Messages, see PARAMETER RANGES & MODULATION for generators:

pitch_range

pitch_rand_type

Example:

pitch_range -12 12 (±1 octave)

Mode 3 - SCALE

Speed parameter = base pitch.

Random note from scale/list of semitones (quantized pitch).

speed_range bypassed.

Formula: final_speed = speed * 2^(scale_note/12)

pitch_scale …

Example:

pitch_scale 0 2 4 5 7 9 11 (major scale)

Up to 128 notes per scale.

Mode 4 - MIDI

Speed parameter = base pitch (assumed C4 = 60).

MIDI note input transposes from base.

speed_range bypassed.

Formula: final_speed = speed * 2^((midi_note - 60)/12)

MIDI input via inlet 19 (1-127, 0 = inactive)

Final speed clamped: -4.0 to 4.0 (±48 semitones / 4 octaves)

MIDI mode “latches” to the last valid note until pitch_mode changes. Switching back to pitch_mode 4 will resume with this latched note.

DELAY

Delay effect applied to the mixed grain output before recording. Three modes available: DD-4 (analog-style), Bencina (pitch-preserving granular), and Stut (quantized rhythmic). All modes share a 9.5-second stereo circular buffer (sized to the host rate) and common control parameters (time, feedback, tone, mix). Mode-specific parameters are set via messages.

Parameters

gdelay_time Set delay time.

Range: 0.0 to 9.5 seconds

Clamped to bounds.

Smoothed internally to prevent clicks during time changes. Smoothing coefficient: 0.001 (approximately 20ms transition at 48kHz).

Signal inlet 10: updates if value in range (0.0, 10.0], else keeps current value.

gdelay_feed Set feedback amount.

Range: 0.0 to 1.0

0.0 = single echo

1.0 = infinite regeneration

Clamped to bounds.

Feedback applied after tone filter in feedback loop (DD-4 and Bencina modes only; inactive in Stut mode).

Signal inlet 11: updates if value in range (0.0, 1.0], else keeps current value.

gdelay_tone Set tone character via one-poleIIR low-pass filter in feedback loop (DD-4 and Bencina modes only; inactive in Stut mode).

Range: 0.0 to 1.0

0.0 = dark, heavily filtered (analog-style decay)

1.0 = bright, no filtering (digital-style)

Clamped to bounds.

Filter coefficient applied directly: filtered = tone × delayed + (1 - tone) × lpf_state

Higher values pass more high frequencies. Lower values create darker, warmer repeats.

Signal inlet 12: In DD-4/Bencina modes, updates feedback if value in range (0.0, 1.0]. In Stut mode (delay_mode 2), controls stut_reduction (gain decay per repeat, range 0.0-1.0).

gdelay_mix Set dry/wet mix.

Range: 0.0 to 1.0

0.0 = 100% dry (grain output only, no delay)

1.0 = 100% wet (delayed signal only)

Clamped to bounds.

Output calculation: output = dry × (1 - mix) + wet × mix

Signal inlet 13: In DD-4/Bencina modes, updates tone if value in range (0.0, 1.0]. In Stut mode (delay_mode 2), controls stut_spacing (repetition spacing in ms, range 1.0-5000.0).

gdelay_clear Clear delay buffers and filterstate.

Zeros buffer_left, buffer_right, lpf_state_left, lpf_state_right.

Immediate silence in feedback loop.

Delay Modes

delay_mode <0|1|2> Select delay algorithm.

DD-4 Mode (delay_mode 0)

Default mode. Classic analog-style delay with a single read head at a fixed distance behind the write head. Feedback and tone filter operate in the feedback loop.

Architecture: input + feedback is written to the buffer. A single read head reads at write_pos minus delay_time with linear interpolation. The delayed signal passes through a one-pole IIR low-pass filter (the tone control), then feedback is calculated from the filtered output.

Active inlets: all four (time, feedback, tone, mix).

Delay time is smoothed internally (coefficient 0.001, approximately 20ms transition at 48kHz) to prevent clicks during time changes.

Bencina Mode (delay_mode 1)

Pitch-preserving granular delay based on the Ross Bencina architecture. Instead of a single read head, spawns up to 32 micro-grains inside the delay buffer. Each grain stores a relative offset from the write head (a “moving tap”), so the grain follows the write head at a fixed distance. This is what preserves pitch — the grain envelope re-synthesizes the delayed audio independently of delay time changes.

Architecture: grains are triggered continuously at a fixed interval (bencina_iot). Each grain reads from write_pos minus its stored offset, applies the main envelope shape as a window, and uses constant-power panning. The sum of all active grains passes through the tone filter, and the filtered output is fed back into the buffer before the write head advances. This true feedback creates recursive granulation — delayed grains get re-granulated — producing shimmer and evolving textures at higher feedback values.

Active inlets: all four (time, feedback, tone, mix). Time controls the distance of the moving tap from the write head. Feedback and tone operate in the grain output path.

bencina_iot Spacing between grain triggers (milliseconds). Controls grain density.

Range: 1.0 to 1000.0 ms. Default: 50.0 ms (20 grains/sec).

Lower values = denser cloud, more overlap between grains. For rhythmic effects, set to a musical subdivision (e.g. 125ms = 1/8 note at 120 BPM).

bencina_grainsize Individual grain length (seconds).

Range: 0.001 to 2.0 seconds. Default: 0.1 seconds (100ms).

Longer grains = smoother, more blurred output. Shorter grains = more granular texture. Applied to future grains only (existing grains keep their size).

bencina_wrap <0|1> Buffer wrap mode.

0 = global (default): grains can read from any part of the buffer, creating layered textures from across the recording. 1 = splice: grains wrap within current splice boundaries, preventing cross-splice bleed.

Applied to all existing and future grains.

bencina_clear Clear all active bencina grains. Immediate silence. Resets active grain count to 0.

Stut Mode (delay_mode 2)

Quantized rhythmic delay. Unlike DD-4 and Bencina which run continuously, Stut is event-triggered via the stut message. When triggered, it captures the current buffer write position and schedules N repetitions at fixed time intervals, each with exponentially decaying gain.

Architecture: the stut message captures an absolute buffer position at the moment of triggering. It then schedules up to 16 grains, each with a countdown timer (i × spacing) and a gain value (gain_reduction ^ i). During processing, each grain counts down; when its timer reaches zero it emits a single-sample pulse from the captured position, scaled by its gain. This produces discrete rhythmic echoes of the captured moment.

Active inlets: time, mix, and — in Stut mode — inlets 12 and 13 are repurposed. Inlet 12 controls stut_reduction (gain decay per repeat, 0.0-1.0) instead of feedback. Inlet 13 controls stut_spacing (repetition spacing in ms, 1.0-5000.0) instead of tone. The stut grains read directly from captured buffer positions with per-repetition gain decay — there is no feedback loop or tone filter in the stut signal path. The mix parameter controls dry/wet blend as usual. Modulation ranges (gdelay_feed range, gdelay_tone range) also route to stut_reduction and stut_spacing respectively when in Stut mode.

stut Trigger a stut sequence. Captures the current splice boundaries and write position. Schedules repetitions immediately. If delay quantization is active (delay_quant > 0, BPM running), uses the quantized grid spacing instead of stut_spacing.

stut_reps Number of repetitions per trigger.

Range: 1 to 16. Default: 4.

stut_reduction Gain reduction factor per repeat. Each repetition's gain = reduction ^ repetition_number.

Range: 0.0 to 1.0. Default: 0.5.

0.0 = maximum reduction (only the first repeat is audible). 0.5 = each repeat is half the volume of the previous. 1.0 = no reduction (all repeats at full volume).

Example with 4 reps, reduction 0.5: gains are [1.0, 0.5, 0.25, 0.125].

stut_spacing Base spacing between repetitions (milliseconds).

Range: 1.0 to 5000.0 ms. Default: 62.5 ms (1/16 note at 120 BPM).

Overridden by delay quantization grid when active (see below).

Stut and Delay Quantization

The stut trigger integrates with the delay quantization system. When all three conditions are met — delay_quant amount > 0, BPM > 1.0, and clock running — the stut trigger uses delay_quant_grid_ms as the spacing between repetitions instead of stut_spacing. This locks stut repeats to the tempo grid.

Example: at 120 BPM with delay_quantize 16 and delay_quant 1.0, stut repeats are spaced at 125ms (1/16 note). Changing delay_quantize to 8 spaces them at 250ms (1/8 note).

Note: delay quantization also affects DD-4 delay time (blending toward the grid), but does not affect Bencina grain spacing. Use bencina_iot directly for Bencina timing.

Per-Mode Parameter Summary

DD-4 Bencina Stut gdelay_time Active Active Inactive* gdelay_feed Active Active → stut_reduction gdelay_tone Active Active → stut_spacing gdelay_mix Active Active Active delay_quant Blends time No effect Overrides spacing Triggering Continuous Continuous Event (stut msg) Read pattern Fixed offset Moving tap Captured position Max voices 1 32 16

- Stut writes input to the shared buffer using the write head, but stut grains read from captured absolute positions. The delay_time parameter does not affect stut grain playback positions.

All three modes share the circular delay buffer and write head. Mode-specific parameters only affect their respective mode. Switching modes via delay_mode takes effect immediately.

FOG (SPECTRAL EFFECT)

FFT-based spectral processing applied to the grain output. Combines horizontal smear (frequency-axis magnitude averaging) and vertical specmagfilter (time-axis resonant filtering of magnitude and phase) to create spectral fog — timbral sustain, blurring, and ghostly persistence.

Signal chain: FFT → polar conversion → smear → specmagfilter → IFFT

Fog is a monitoring effect — it is not recorded into the reel. Applied after delay and before distortion in the effects chain.

Signal inlet 15: fog mix (0.0-1.0). 0.0 = dry bypass (no FFT processing). 1.0 = full wet (100% spectral effect). Equal-power crossfade between dry and wet.

Smear (Horizontal)

Frequency-axis magnitude averaging. Blurs the spectral content across neighboring bins, creating a diffuse, hazy timbre.

fog_smear_bins Number of neighbor bins to average on each side.

Range: 0 to 32. Default: 8.

Higher values = wider blur, more diffuse timbre. 0 = no smearing.

fog_smear_enable <0|1> Enable/disable smear processing.

fog_smear_onset_curve <0|1|2> How smear amount scales with fog mix. 0 = linear, 1 = exponential (x^2, delayed onset), 2 = logarithmic (sqrt, early onset). Default: 2.

fog_smear_onset_amount <0-1> Maximum smear depth at mix=1.0. Default: 1.0.

SpecMagFilter (Vertical)

Time-axis resonant filtering. Lowpass filters magnitude and phase per bin across consecutive FFT frames, creating temporal persistence — spectral content decays slowly rather than updating instantly.

fog_mag_cutoff Magnitude filter cutoff frequency. Controls how quickly magnitudes track changes.

Range: 0.1 to 20.0 Hz (relative to FFT hop rate). Default: 2.0 Hz.

Lower values = slower tracking, longer spectral sustain. Higher values = faster response.

fog_mag_resonance Magnitude filter resonance (Q factor). Adds a peak at the cutoff frequency, emphasizing the rate of spectral change.

Range: 0.1 to 10.0. Default: 1.0.

Higher values create a more pronounced, ringy spectral persistence. Soft-limited via tanh to prevent magnitude explosion.

fog_phase_cutoff Phase lowpass cutoff. Controls smoothing of phase evolution between frames.

Range: 0.1 to 20.0 Hz. Default: 2.0 Hz.

Lower values freeze phase relationships, creating a static spectral snapshot.

fog_specmagfilter_enable <0|1> Enable/disable temporal filtering.

fog_specmagfilter_onset_curve <0|1|2> How filter depth scales with fog mix. 0 = linear, 1 = exponential, 2 = logarithmic (default).

fog_specmagfilter_onset_amount <0-1> Maximum filter depth at mix=1.0. Default: 1.0.

Stereo Filter Mode

fog_stereo_filter_mode <0|1> Select stereo processing strategy.

0 (default): Shared state. Left and right channels share the same per-bin filter state arrays. Creates a diffuse, mono-ish spectral character — both channels converge toward the same spectral evolution.

1: Independent. Each channel has its own per-bin state arrays. Left and right channels evolve independently, preserving true stereo spectral separation.

Per-Grain Fog Mode

fog_position <0|1> Select fog routing.

0: Per-grain. Each grain is assigned to one of N independent fog slots via round-robin at trigger time. Grains accumulate into per-slot buffers, then each slot runs its own FFT pipeline. Different grains develop independent spectral evolution — some fogging faster, some slower — creating richer, more complex textures than global processing.

1 (default): Post-mix. Single fog instance processes the mixed grain output. Existing behavior.

Pool size is configurable via ligase.conf (fog_pool_size, 1-8 slots, default 4). Each slot costs ~76KB.

Configuration (ligase.conf)

fft_size = 1024 FFT window size: 512, 1024 (default), or 2048. 512 = lower latency, less frequency resolution. 2048 = highest resolution, higher latency.

overlap_factor = 4 Overlap factor: 2, 4 (default), or 8. Higher = smoother temporal evolution, more CPU.

fog_pool_size = 4 Per-grain fog slots: 1-8, default 4. ~76KB per slot.

DISTORTION

Multi-mode waveshaping with pre/post filters and oversampling. Two position modes: per-grain (experimental) or post-mix (default). In post-mix mode, distortion is applied only AFTER recording as a monitoring effect.

Control

distortion_enable 0|1 Enable/disable distortion processing.

0 = bypass (passthrough)

1 = enabled

Signal inlet 14: updates if value in range (0.0, 1.0], else keeps current value.

distortion Set distortion drive amount.

Range: 0.0 to 1.0

Maps to current_drive via lookup curve.

0.0 = clean (minimal drive)

1.0 = saturated (maximum drive)

distortion_position 0|1 Set distortion placement in signal chain.

Mode 0: Per-grain (experimental)

Applied to each grain sample during synthesis, before envelope. No oversampling. Uses grain_distortion_process_sample().

Signal chain: read → distort → envelope → pan → mix

Distortion IS recorded in this mode (affects grain output before recording).

Mode 1: Post-mix (default)

Applied to final output AFTER all recording operations.

Uses oversampling. Uses grain_distortion_process_block().

Signal chain: grains → delay → [RECORDING] → fog → distortion → Moog → dac~

Distortion NOT recorded (monitoring/output only).

This placement prevents distortion artifacts from accumulating in overdub recordings.

Default: 1 (post-mix)

distortion_oversample_factor 1|2|4|8 Set oversampling multiplier for post-mix mode. Valid values: 1, 2, 4, 8 ?

Posts error if invalid value provided.

Higher values reduce aliasing at cost of CPU.

4x provides good quality/performance balance.

Per-grain mode ignores this setting (no oversampling).

Default: 4

Waveshaper Modes

dist_waveshaper_mode 0|1|2|3|4 Select waveshaping algorithm.

Mode 0: tanh (default)

Pure hyperbolic tangent.

Aggressive odd harmonics. Warm, tube-like saturation.

Mode 1: arctan

Pure arctangent.

Smooth odd harmonics. Gentler than tanh.

Mode 2: asymmetric

Different drive for positive/negative cycles.

Generates even harmonics.

Uses drive_pos and drive_neg parameters.

Mode 3: blend

Morph between tanh and arctan.

Uses dist_curve_blend parameter (0.0 = tanh, 1.0 = arctan).

Mode 4: polynomial

User-defined polynomial: output = $c1 \times x + c2 \times x^2 + c3 \times x^3$

Uses poly_c1, poly_c2, poly_c3 coefficients.

Precise harmonic control.

Default: 0 (tanh)

dist_curve_blend Blend amount for Mode 3.

Range: 0.0 to 1.0

0.0 = full tanh (aggressive)

1.0 = full arctan (smooth)

dist_drive_pos Positive cycle drive for Mode2. Range: 1.0 to 20.0

Higher values = more saturation on positive peaks.

dist_drive_neg Negative cycle drive for Mode2.

Range: 1.0 to 20.0

Higher values = more saturation on negative peaks.

dist_poly_c1 Linear coefficient for Mode 4.

Range: -10.0 to 10.0

Controls fundamental and odd harmonics.

Default: 1.0 (linear passthrough)

dist_poly_c2 Quadratic coefficient for Mode 4.

Range: -10.0 to 10.0

Controls even harmonics (asymmetry).

dist_poly_c3 Cubic coefficient for Mode 4.

Range: -10.0 to 10.0

Controls additional odd harmonics.

Pre-emphasis / De-emphasis (Inner Filters)

Frequency-dependent pre-distortion boost with matched post-distortion cut. Affects harmonic content without changing overall tonal balance.

dist_emphasis_mode 0|1 Set pre-emphasis filter type.

0 = HP (high-pass): boost highs before distortion → brighter saturation

1 = LP (low-pass): boost lows before distortion → darker saturation

Argument >= 0.5 selects LP, < 0.5 selects HP.

Default: 0 (HP mode)

dist_emphasis_freq Set emphasis corner frequency. Range: 100 to 5000 Hz

Higher freq = more extreme emphasis effect.

dist_pregain Set pre-distortion gain stage.

Range: 0.1 to 10.0

Applied before emphasis filters and waveshaper.

1.0 = unity1.0 = drive harder

< 1.0 = soften

Outer Filters

Applied outside emphasis/waveshaper core. Shape frequency response without affecting emphasis balance.

distortion_pre_hp_freq Pre-tanh highpass frequency.

Range: 30 to 500 Hz

1-pole filter (6dB/octave).

Removes DC and low mud before saturation.

distortion_pre_hp_mix Pre-tanh highpass dry/wet.

Range: 0.0 to 1.0

0.0 = full passthrough (no filtering)

1.0 = full filtered

distortion_post_lp_freq Post-tanh lowpassfrequency.

Range: 2400 to 10000 Hz

2-pole Butterworth (12dB/octave).

Tames high-frequency harmonics after saturation.

distortion_post_lp_mix Post-tanh lowpass dry/wet.

Range: 0.0 to 1.0

0.0 = full bright (no filtering)

1.0 = full dark

distortion_notch_freq Notch filter centerfrequency.

Range: 20 Hz to (sample_rate/2 - 100) Hz

Bandstop filter for targeted frequency removal.

distortion_notch_bw Notch filter bandwidth.

Range: 10 to 5000 Hz

Narrower = more surgical cut.

distortion_notch_mix Notch filter dry/wet.

Range: 0.0 to 1.0

0.0 = inactive (no notching)

1.0 = full notch applied

Default: 0.0 (inactive)

MOOG LADDER FILTER

Classic 4-pole (24dB/octave) Moog ladder lowpass filter. Four cascaded one-pole stages with resonance feedback. Filter is applied only AFTER recording as a monitoring effect.

Parameters

moog_cutoff Set filter cutoff frequency.

Range: 20 to 20000 Hz

Clamped to Nyquist frequency (sample_rate / 2).

Formula: $f = 2 \times \sin(\pi \times fc / fs)$

Cutoff coefficient clamped to [0.0, 1.0] for stability.

Signal inlet 16: updates if value in range [20.0, 20000.0], else keeps current value.

moog_resonance Set resonance feedback amount.

Range: 0.0 to 4.0

0.0 = no resonance (flat lowpass)

3.5+ = self-oscillation (pitched sine tone at cutoff frequency)

4.0 = maximum feedback

Feedback formula: $fb = resonance \times (1.0 + 0.5 \times f^5)$

Nonlinear compensation prevents resonance drop at high cutoff frequencies.

Signal inlet 16: updates if value in range [0.0, 4.0], else keeps current value.

moog_mix Set dry/wet mix.

Range: 0.0 to 1.0

0.0 = 100% dry (bypass, no filtering)

1.0 = 100% wet (fully filtered)

Output: $output = dry \times (1 - mix) + wet \times mix$

Signal inlet 17: updates if value in range [0.0, 1.0], else keeps current value.

moog_enable 0|1 Enable/disable filter processing.

0 = bypass (passthrough)

1 = enabled

When disabled or mix = 0.0, filter bypasses without processing.

PARAMETER RANGES & MODULATION

Per-grain parameter variation system. Each parameter can vary within a specified range driven by one of nine generator types (rand, Perlin 1D/2D, Lorenz, N-body, sphere, saw, sine, square — four instances each). ~37 parameters across grain synthesis, timing/playback, delay, distortion, fog, and the modulation outlets are modulatable.

Modulation Sources

Five generator types, each with 4 independent instances (1-4):

- RAND_TYPE_RAND

Basic pseudo-random. Seeded per instance. Uniform distribution [0.0, 1.0]. Fast, uncorrelated white noise variation.

- RAND_TYPE_PERLIN_1D

1D Perlin noise. Smooth continuous modulation. Coordinate advances by iot × noise_frequency_scale per grain. Decorrelated via instance offsets: [0, 1000, 2003, 3001]. Returns normalized [0.0, 1.0].

- RAND_TYPE_PERLIN_2D

2D Perlin noise. X-coordinate advances per grain, Y-coordinate = base parameter value.

Coupling between parameter value and modulation depth. Decorrelated via instance offsets: [0, 5000, 10007, 15013]. Returns normalized [0.0, 1.0].

- RAND_TYPE_LORENZ

Lorenz attractor. Bounded chaotic motion. Evolves via update iterations: int(iot × 100 × noise_frequency_scale). Capped 1-50 iterations per grain for stability. Output axis rotates per instance: 0=X, 1=Y, 2=Z (mod 3). Different initial positions per instance for decorrelation. Returns normalized [0.0, 1.0].

- RAND_TYPE_NBODY

N-body gravitational simulation (3-body problem). Bounded chaotic orbital motion.

Evolves via update iterations like Lorenz. Output mode selectable per instance (0-10):

Body positions, velocities, distances, angular momentum, total energy. Returns normalized [0.0, 1.0].

- RAND_TYPE_SPHERE

3D sphere physics simulation (STK-based). A sphere bounces within a bounded space with configurable damping and elasticity. Each instance can be “kicked” with a velocity impulse. Output mode selectable per instance (0-6): 0=Position X, 1=Position Y, 2=Position Z, 3=Velocity X, 4=Velocity Y, 5=Velocity Z, 6=Velocity Magnitude. Returns normalized [0.0, 1.0].

sphere_kick Apply velocity impulse to sphere instance. Creates sudden, decaying motion patterns.

sphere_damping <0-1> Set damping coefficient. Higher values = faster energy decay.

sphere_elasticity <0-1> Set bounce elasticity. 1.0 = perfect elastic bounce. Lower values absorb energy on wall contact.

sphere_reset Reset sphere to initial position and zero velocity.

sphere_mode <0-6> Set output mode. 0-2 = position axes, 3-5 = velocity axes, 6 = velocity magnitude.

- RAND_TYPE_SAW

Sawtooth wave LFO. Unipolar ramp from 0.0 to 1.0. Phase advances by iot × noise_frequency_scale per grain trigger. Creates predictable linear sweep patterns for modulation.

- RAND_TYPE_SINE

Sine wave LFO. Unipolar 0.0 to 1.0 (mapped from bipolar sine). Phase advances by iot × noise_frequency_scale per grain trigger. Creates smooth, predictable oscillating modulation.

- RAND_TYPE_SQUARE

Square wave LFO. Bipolar switching: 0.0 below 50% duty cycle, 1.0 above. Phase advances by iot × noise_frequency_scale per grain trigger. Creates alternating on/off modulation patterns.

Modulatable Parameters Include:

- Grain synthesis (sampled per grain trigger):

speed, grainsize, grainstart, amplitude, pan, maxgrains, env_skew

- Timing / playback (sampled per DSP block):

iot (interonset time), scanrate, organize (splice select), sos (sound-on-sound)

- Effects (sampled per DSP block):

gdelay (delay time), gdelay_feed, gdelay_tone, gdelay_mix, moog_cutoff, moog_resonance, moog_mix

- Distortion:

distortion (intensity), dist_emphasis_freq, dist_pregain, dist_curve_blend, dist_drive_pos, dist_drive_neg, dist_poly_c1, dist_poly_c2, dist_poly_c3

- Fog (sampled per DSP block):

fog_mix, fog_smear_bins, fog_smear_onset, fog_mag_cutoff, fog_mag_resonance, fog_phase_cutoff, fog_smf_onset

- Stut (sampled per DSP block):

stut_reps

- Bencina/PPG (sampled per DSP block):

bencina_iot, bencina_grainsize

- Modulation outlets (the four send outlets are themselves modulatable targets):

modout1, modout2, modout3, modout4

Messages

- param_range [max] Set parameter modulation range.
  - Two-argument form (single value): Disables range, sets fixed value. Example: param_range speed 1.0→ speed fixed at 1.0
  - Three-argument form (range): Enables modulation between min and max. Example: param_range speed 0.5 2.0→ speed varies 0.5-2.0
- rand_type [param_name] Set modulation source for parameter(s).
  - type_instance format: {type}_{N} where N = 1-4
  - Types: rand_1…rand_4, perlin_1d_1…perlin_1d_4, perlin_2d_1…perlin_2d_4, lorenz_1…lorenz_4, nbody_1…nbody_4, sphere_1…sphere_4, saw_1…saw_4, sine_1…sine_4, square_1…square_4
  - One-argument form: Sets all parameters to specified source. Example:

rand_type perlin_1d_2

- Two-argument form: Sets single parameter to specified source. Example:

rand_type lorenz_3 amplitude

- noise _freq Set frequency scale for all 4 modulation instances.
  - Range: > 0.0 (1.0 = normal speed)

    - Affects Perlin coordinate advancement and Lorenz/N-body iteration count.

- noise _freq_1
- noise _freq_2
- noise _freq_3
- noise _freq_4

Set frequency scale for specific instance (1-4). Independent control of each modulation instance speed.

Lock

Sends current modulated values on outlet 9 then stops modulation.

param_lock …

Example:

param_range pan 0.2 0.8

rand_type perlin_1d_1 pan

param_lock pan → "ligase~: pan locked at 0.547"

param_lock pan amplitude grainsize → locks all three at current values

Disables range modulation only.

Returns to inlet or last message value.

Without feedback routing:

→ Just stops modulation

→ Returns to inlet/message control

→ Values go to outlet 9 but aren't applied

With feedback routing (outlet 9 → inlet 1):

→ Stops modulation

→ Captures and applies current modulated values

→ "Locks" parameters at modulated state

N-Body Control

N-body instances have additional physics parameters (1-4):

- nbody_epsilon Set softening parameter (0.001-0.1). Prevents singularities during close approaches.
- nbody_damping Set velocity damping (0.0-0.1). Higher values = faster energy loss, more stable orbits.
- nbody_pump Set energy pump interval (velocity increment, iteration interval). Adds velocity periodically to sustain motion
- nbody_G Set gravitational constant(0.1-10.0).
- nbody_mode Set output extraction mode (0-10).

11 output modes per instance:

0: Body 0 X position (instance 1 default)

1: Body 1 Y position (instance 2 default)

2: Body 2 X position (instance 3 default)

3: Distance body 0-1 (instance 4 default)

4: Velocity magnitude body 0

5: Velocity magnitude body 1

6: Velocity magnitude body 2

7: Distance body 0-2

8: Distance body 1-2

9: Angular momentum magnitude

10: Total system energy

Auto-normalizing bounds track min/max values.

Sampling Timing

- Per grain trigger: speed, grainsize, grainstart, amplitude,pan, and maxgrains
- Per DSP block (smooth variation): iot, gdelay, moog_cutoff,moog_resonance, moog_mix.

Modulation Mapping

Output value: min + random_value × (max - min). Linear mapping using normalized random_value [0.0, 1.0] from the generator.

Generator Outlet Sends

The four modulation outlets (modout1–modout4) are first-class modulatable parameters. Configure them with the same unified messages used for any parameter, addressing them by name (modout1… modout4); there are no separate modout_source/modout_range commands:

rand_type modoutN - assign a generator (any of the nine types, instance 1-4) to outlet N param_range modoutN - set the outlet's output range param_slew / param_invert modoutN - smooth / invert the outlet, as for any parameter

Per-Outlet Base Value

Set PERLIN_2D Y-coordinate for parameter decorrelation.

param_base_value

Example:

rand_type perlin_2d_1 modout1

rand_type perlin_2d_1 modout2

param_base_value modout1 0.2

param_base_value modout2 0.8

Result:

modout1 and modout2 use same instance but different Y-coordinates, producing decorrelated patterns.

Notes:

Default: 0.5 for all parameters

Only affects PERLIN_2D (ignored for other generator types)

Enables spatial decorrelation without using multiple instances

Parameter Slewing

Apply exponential smoothing to modulation output.

param_slew

Example:

rand_type rand_1 modout1

param_range modout1 0 1

param_slew modout1 0.9 → Very smooth, slow tracking

param_slew modout1 0.5 → Moderate smoothing

param_slew modout1 0.0 → Instant (no smoothing)

Range:

0.0: Instant response (disabled)

0.5: Moderate smoothing

0.99: Very slow tracking

1.0: Frozen (no updates)

Notes:

Algorithm: smoothed = smoothed × slew + new × (1 slew)

Works with all generator types

Generator Reset

Reset Perlin or Lorenz generators to initial state.

perlin_reset

lorenz_reset

Example:

perlin_reset 1 → Resets Perlin 1D/2D coordinates to 0

lorenz_reset 2 → Resets Lorenz attractor to initial position

nbody_reset 3 → Resets N-body system (existing feature)

Perlin Reset:

Resets noise_1d_coord and noise_2d_coord_x to 0.0

Useful for synchronizing modulation patterns

Lorenz Reset:

Returns x, y, z to initial values (x0, y0, z0)

Each instance has different starting position:

Instance 1: (0.1, 0.0, 0.0)

Instance 2: (5.0, 10.0, 20.0)

Instance 3: (-3.0, -5.0, 10.0)

Instance 4: (8.0, 15.0, 30.0)

N-body Reset:

Resets to hierarchical initial configuration

Each instance has different mass distribution

Notes:

Instance range: 1-4

Does not reset frequency scaling (perlin_frequency)

Does not reset N-body physics parameters (epsilon, damping, etc.)

Defaults

All parameter ranges initialized to: min: 0.0, max: 1.0, rand_type: RAND_TYPE_RAND (instance 1), enabled: 0 (disabled). All noise_frequency_scale defaults to 1.0.

QUERY STATE

Export and inspect ligase~ state via outlet 9.

Useful for creating presets or routing values to other instances:

query → Output single parameter value

get_params → Output all 20 parameter values

Debugging and inspection:

get_inlets → Print inlet guide to console

get_ranges → Output active param_range settings

get_generators → Output active rand_type assignments

get_state → Output everything (params + ranges + generators)

query bpm → returns calculated bpm

get_inlets

Print inlet documentation to console (no outlet).

Output: Console only

Shows current values and message syntax

query

Query single parameter value.

query

20 queryable parameters:

speed, grainsize, grainstart, organize, scanrate, sos, iot, maxgrains, gdelay, gdelay_feed, gdelay_tone, gdelay_mix, distortion, moog_cutoff, moog_resonance, moog_mix, midi, env_skew, amplitude, pan

Per-block parameters - store modulated values when param_range.enabled: gdelay_time, gdelay_feedback, gdelay_tone, gdelay_mix moog_cutoff, moog_resonance, moog_mix organize, sos, env_skew, iot

Per-grain parameters - Sample once per block for query when modulation enabled:

speed, grainsize, grainstart, amplitude, pan, maxgrains, distortion

query and get_params return:

Inlet/ last message value when modulation is disabled

Current modulated value when modulation is active

Locked value when param_lock has been applied

Unconnected inlets preserve stored values if headless mode is enabled

get_params

Export all 20 parameter values.

Fixed order (speed → grainsize → … → pan)

get_ranges

Export active parameter range configurations.

Output format:

param_range

Only outputs enabled ranges

Messages can be sent back to restore range configuration

get_generators

Export active generator assignments.

Output format:

rand_type _

Generator types: rand, perlin_1d, perlin_2d, lorenz, nbody

Instance numbers 1-4

Only outputs modout1-4 (scheduler generators not accessible)

Does NOT export perlin_frequency or nbody settings

get_state

Export complete state (all three subsystems).

Output order: 1. All parameter values (20 messages) 2. All enabled ranges (variable) 3. All generator assignments (variable)

Calls get_params + get_ranges + get_generators

All messages are routable lists

Limitations

No quantization state - timesig/quant settings not exported